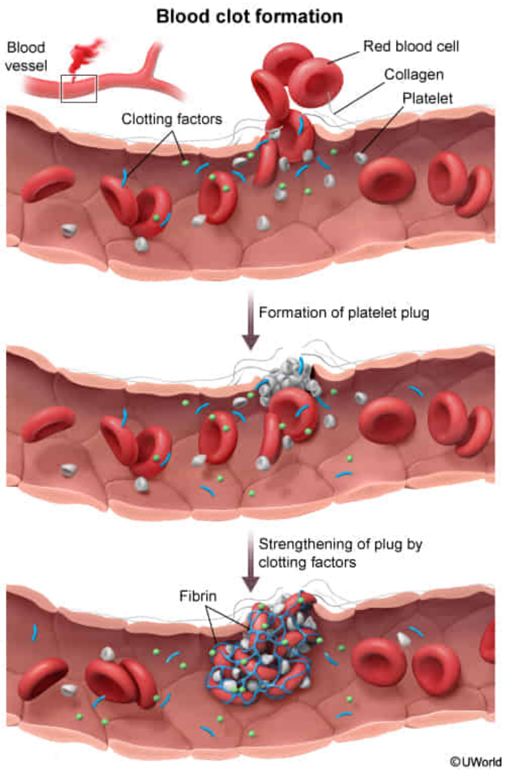
A physician finds that a patient experiences excessive bleeding after sustaining a cut on the arm. The excessive bleeding is LEAST likely to be caused by:

- A. reduced liver function. [27%]
- ✔ B. high numbers of macrophages near the cut. [60%]
- C. reduced production of platelets. [8%]
- D. high levels of serum antibodies against clotting factors. [3%]

Correct
59% answered correctly

Explanation:



A single layer of endothelial cells comprises the inner wall of all blood vessels, forming a barrier that regulates the entry and exit of materials into and out of the bloodstream. In addition, endothelial cells secrete chemical signals that generally prevent blood coagulation (ie, clotting). However, injury to a blood vessel that results in bleeding indicates damage to the endothelium and triggers the following cascade of events:

1. **Formation of the platelet plug**. Endothelial damage exposes connective tissue (ie, collagen fibers) normally present outside the blood vessel. Circulating cell fragments called **platelets** (derived from the bone marrow) readily bind these collagen fibers, aggregating to form a **platelet plug** (clot) that prevents blood flow out of the vessel. In addition, bound platelets and endothelial cells near the site of damage continue to release signals that *enhance* platelet aggregation.
2. **Strengthening of the clot**. Clotting factors (mainly synthesized in the liver) are specialized proteins that become activated in response to platelet aggregation and signaling factors outside the vessel. Activated clotting factors induce processes that lead to the formation of the enzyme **thrombin**. Thrombin induces protein strands (ie, **fibrin**) to form an adhesive mesh-like structure over the platelet plug, reinforcing the clot.

In this scenario, a patient with a cut experiences excessive bleeding, indicating an **impaired ability to form clots** and prevent blood loss from damaged vessels. This symptom could have several causes, including *reduced* liver function, which would likely *decrease* clotting factor production **(Choice A)**, and *reduced* platelet production **(Choice C)**. In addition, because antibodies mark cellular materials for destruction by immune cells, heightened levels of serum antibodies *against* clotting factors would promote destruction of these factors, *impairing* clot formation **(Choice D)**.

Macrophages, phagocytic immune cells, engulf and destroy foreign pathogens and damaged cells. These cells may destroy pathogens entering the body through the cut and can contribute to clot formation (eg, by secreting procoagulant molecules). Therefore, **macrophages would not inhibit clot formation** and cause the excessive bleeding observed in the patient.

Educational objective:
Platelets and specialized proteins (ie, clotting factors) work synergistically in response to blood vessel damage, binding to the damaged portion of the vessel and forming a blockage that prevents blood loss.

Time spent:0 sec
Subject:
Biology

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